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Comparison of gluteal and hamstring activation during five commonly used plyometric exercises

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ABSTRACT

Background: Anterior cruciate ligament injuries occur frequently in athletics, and anterior cruciate ligament injury prevention programs may decrease injury risk. However, previous prevention programs that include plyometrics use a variety of exercises with little justification of exercise inclusion. Because gluteal and hamstring activation is thought to be important for preventing knee injuries, the purpose of this study was to determine which commonly used plyometric exercises produce the greatest activation of the gluteals and hamstrings.
Methods: EMG (Electromyography) amplitudes of the hamstring and gluteal muscles during preparatory and loading phases of landing were recorded in 41 subjects during 5 commonly used plyometric exercises. Repeated measures ANOVAs (Analysis of Variance) were used on 36 subjects to examine differences in muscle activation.
Findings: Differences in hamstring ($P < .01$) and gluteal ($P < .01$) activities were identified across exercises during the preparatory and landing phases. The single-leg sagittal plane hurdle hops produced the greatest gluteal and hamstring activity in both phases. The 180° jumps did not produce significantly greater gluteal or hamstring activity than any other exercise.
Interpretation: Single-leg sagittal plane hurdle hops may be the most effective exercise to activate the gluteals and hamstrings and may be important to include in anterior cruciate ligament injury prevention programs, given the importance of these muscles for limiting valgus loading of the knee. Because 180° jumps do not produce greater gluteal and hamstring activation than other plyometric exercises, their removal from injury prevention programs may be warranted without affecting program efficacy.

1. Introduction

While considerable debate exists regarding the explicit mechanism(s) of anterior cruciate ligament (ACL) injury, knee valgus motion has received considerable attention in the literature as an ACL loading mechanism. Hewett et al. (2005) reported that peak knee valgus angle during a jump landing prospectively predicted ACL injury risk in adolescent females. Therefore, limiting knee valgus motion during training and competition may reduce ACL injury risk. In the closed kinetic chain (e.g., cutting, pivoting, landing), knee valgus motion results from a combination of movements including hip adduction and internal rotation. Because the gluteal muscles (gluteus maximus and medius) provide eccentric resistance to these motions (Neumann, 2010), gluteal strengthening has been advocated in efforts to limit knee valgus (Willson et al., 2006). However, strengthening the gluteal muscles does not affect the amount of knee valgus exhibited during dynamic tasks (Herman et al., 2008).

Dynamic tasks such as landing do not require maximal effort; thus a muscle's activation level may provide a better indication of its role in controlling kinematics than its strength (Distefano et al., 2009; Zazulak et al., 2005). Greater gluteal activity limits knee valgus motion by controlling hip adduction and hip internal rotation (Hollman et al., 2009; Patrek et al., 2011; Zazulak et al., 2005). Hamstring activation, especially on the medial side, may also limit knee valgus motion by controlling frontal plane knee motion (Lloyd et al., 2005). Subjects who exhibit earlier and greater medial hamstring activation during both the preparatory and loading phases of landing display lesser knee valgus moments than those with lesser medial hamstring activity (McLean et al., 2010, Palmieri-Smith et al., 2008) (Fig. 1). Previous research has also shown that the activity of the medial hamstring muscles limits knee valgus loading during a variety of tasks including controlled perturbations (Dhaher et al., 2005; Zhang and Wang, 2001) and during static (Claiborne et al., 2006) and dynamic (Besier et al., 2003) tasks. Therefore, enhancing gluteal and hamstring

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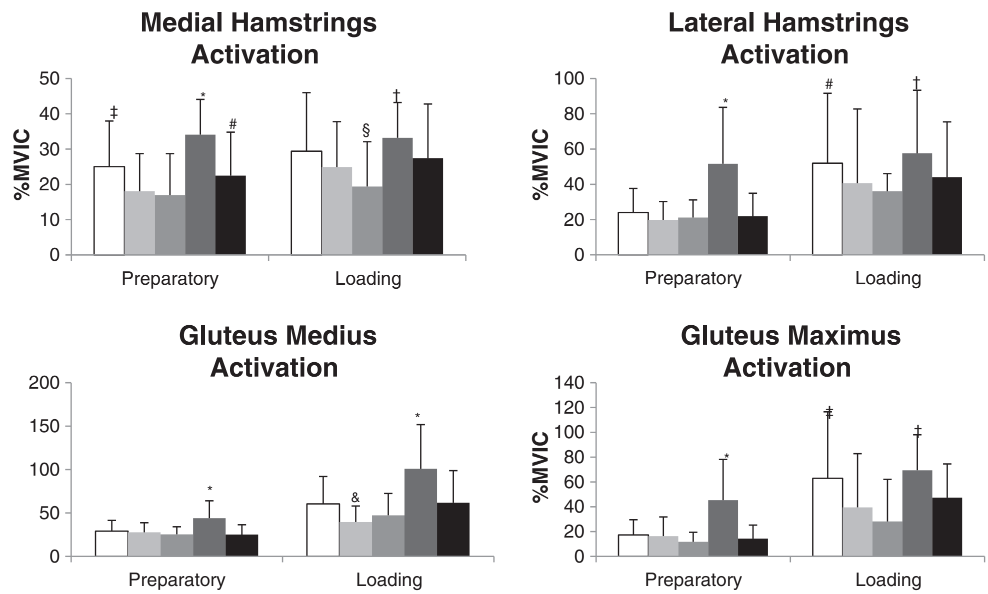

Fig. 1. Comparisons of average activation of all muscles between plyometric exercises.

□ dlsag = double-leg sagittal plane hurdle hops
frontal = frontal plane hurdle hops
180 = 180° jumps
slsag = single-leg sagittal plane hurdle hops
■ splitsquat = split squat jumps. dlsag = double leg sagittal plane hurdle hops; frontal = frontal plane hurdle hops; 180 = 180° hops; slsag = single leg sagittal plane hurdle hops; splitsquat = split squat jumps. *Significantly greater than dlsag, frontal, 180, and splitsquat of the same phase. †Significantly greater than frontal, 180, and splitsquat of the same phase. ‡Significantly greater than frontal and 180 of the same phase. #Significantly greater than 180 of the same phase. §Significantly less than loading phase dlsag, frontal, slsag, and splitsquat of the same phase. &Significantly less than loading phase dlsag, slsag, and splitsquat of the same phase.

activation rather than enhancing strength may improve ACL injury prevention efforts.

Although ACL prevention programs commonly use several exercise modalities, plyometric exercises seem to be important elements of ACL injury prevention programs due to the fact that they may decrease knee valgus moments and angles if completed correctly (Markovic and Mikulic, 2010; Myer et al., 2005, 2006b). Plyometric exercises may be able to improve landing and cutting biomechanics by increasing muscle activation and improving neuromuscular effectiveness (Potteiger et al., 2005). However, the literature regarding the efficacy of these prevention programs is equivocal, as some investigations have reported significant decreases in ACL injury risk (Hewett et al., 1999; Mandelbaum et al., 2005) while others report a little-to-no effect (Myklebust et al., 2003; Pfeiffer et al., 2006; Steffen et al., 2008).

The discrepancies between prevention programs may be explained by the wide variety of exercises included in ACL prevention programs. A program which included plyometric exercises that were designed to target the gluteal and hamstring musculature resulted in a significant reduction in ACL injury risk, but the authors noted that further analysis was needed to ensure that the exercises provided protection to the ACL (Mandelbaum et al., 2005). Stepping and cutting exercises performed in the frontal plane exercises have been shown to recruit the medial hamstrings and gluteals to a greater extent than similar exercises performed in sagittal plane (Houck, 2003; Mercer et al., 2009). Single-leg squatting also produces greater gluteal activity than double-leg squatting (Lubahn et al., 2011). However, it is unknown whether more physically demanding exercises, such as plyometrics, performed in different planes also produce differences in hamstring and gluteal activities. Given the perceived importance of gluteal and hamstring muscle activation in ACL injury prevention, the purpose of this study was to compare the activation of these muscles between five commonly used plyometric exercises. It was hypothesized that exercises involving frontal plane motions would generate greater hamstring/gluteal activation compared to sagittal plane exercises and that single-leg landings would yield greater activation than double-leg landings.

2. Methods

2.1. Subjects

Forty-one volunteers (20 males, 21 females; mass = 70.3[13.4] kg; height = 70.32[10.2] cm) between the ages of 18 and 30 years participated in this investigation. Subjects were physically active (participation in physical activity at least 30 min per day, 3 days per week) and had no history of lower extremity or back injury within the 6 months prior to participation or surgery in either lower extremity. Subjects read and signed an informed consent document approved by the University's Institutional Review board and were asked to refrain from resistance training within the 48 h prior to testing.

2.2. Experimental procedures

Prior to testing, differential surface electromyography (EMG) electrodes (DelSys, Inc., Boston, MA, USA: interelectrode distance 10 mm; amplification factor 1000 (20–450 Hz); CMMR @ 60 Hz > 80 dB; input impedance > 10^{15}//0.2 Ω//pF) were placed over the gluteus maximus, gluteus medius, medial hamstrings, and lateral hamstrings of the dominant limb (the limb used to kick a ball for maximum distance) approximately parallel to the muscle fiber orientation in accordance with SENIAM guidelines (Hermens et al., 2000). Electrodes were secured using a double-sided adhesive skin interface to measure mean EMG amplitudes during the preparatory and landing phases of each muscle during each plyometric exercise. These amplitudes were normalized to maximum voluntary isometric contractions (MVICs) during hip extension, hip abduction, and knee flexion (Hislop and Montgomery, 1995) to provide normalization values for the gluteus maximus, gluteus medius, and hamstrings, respectively (Distefano et al., 2009). Electrodes and leads were secured to the skin using tape to minimize motion artifact. Electrode sites were shaved if necessary, abraded, and cleaned with isopropyl alcohol to maximize the electrode adherence to the skin and minimize skin impedance. Electrode placements were confirmed via isometric contractions against manual resistance.

Table 1
Plyometric exercise descriptions.

Exercise	Description
180° jump (transverse plane) Chimera et al. (2004), Hewett et al. (1996), Irmischer et al. (2004), Myer et al. (2006a), Myklebust et al. (2003), Wilkerson et al. (2004)	The subject was instructed to stand with the non-dominant foot on the force plate and the dominant foot off the force plate. The subject then jumped while performing a 180° turn in the transverse plane toward the non-dominant shoulder, landing with the non-dominant foot off the force plate and the dominant foot on the force plate. On the next beat of the metronome, the subject jumped and completed another 180° turn over the dominant shoulder, returning to the starting position.
Frontal plane hurdle hop Chimera et al. (2004), Irmischer et al. (2004), Lephart et al. (2005), Myer et al. (2006a), Pasanen et al. (2008), Steffen et al. (2008), Wilkerson et al. (2004)	The subject was instructed to start standing, feet straddling a line, with the dominant foot at a distance of 50% of the subject's height from the center of one of the force plates and the non-dominant leg closest to the force plate. A 10.16 cm tall hurdle was placed halfway between the subject's dominant leg and the center of the force plate. The subject then jumped laterally over the hurdle toward his/her non-dominant leg and landed with the dominant foot on the force plate and the non-dominant foot off of the force plate. On the next beat of the metronome, the subject jumped in the opposite direction over the hurdle and returned to the initial starting position.
Double-leg sagittal plane hurdle hop Arabatzi et al. (2010), Hewett et al. (1996), Lephart et al. (2005), Myer et al. (2006a), Myklebust et al. (2003), Pasanen et al. (2008), Steffen et al. (2008), Wilkerson et al. (2004)	The subject was instructed to start standing behind a line with the feet at a distance 50% of his/her height from the center of the force plates. A 10.16 cm tall hurdle was placed halfway between the subject's feet and the center of the force plates. The subject jumped forward over the hurdle in the sagittal plane and landed with the dominant foot on the force plate and the non-dominant foot off the force plate. On the next beat of the metronome, the subject jumped backward over the hurdle and return to the initial starting position.
Single-leg sagittal plane hurdle hop Arabatzi et al. (2010), Hewett et al. (2006), Irmischer et al. (2004), Lephart et al. (2005), Pasanen et al. (2008)	The subject was instructed to start standing on the dominant foot behind a line a distance 30% of his/her height from the center of the force plate. A 10.16 cm tall hurdle was placed halfway between the subject's feet and the center of the force plate. The subject jumped forward over the hurdle in the sagittal plane. The subject landed with the dominant foot on the force plate and was not allowed to touch down with the non-dominant leg. On the next beat of the metronome, the subject jumped backward over the hurdle and return to the initial starting position.
Split squat jump (sagittal plane) Chimera et al. (2004), Hewett et al. (1996), Lephart et al. (2005), Pasanen et al. (2008), Wilkerson et al. (2004)	The subject was instructed to begin in a lunge position with the non-dominant leg immediately lateral to the force plate and the dominant limb behind the force plate. The subject jumped in the air while moving the non-dominant limb backward and immediately the dominant limb forward onto the force plate, landing in a lunge position. On the next beat of the metronome, the subject jumped as high as possible and switched the legs back to the starting position.

Before the testing session, subjects completed a five minute warm-up on a stationary cycle ergometer at a self-selected pace. The primary researcher then demonstrated and explained the plyometric exercises (180° jump, frontal plane hurdle hop, double leg sagittal plane hurdle hop, single leg sagittal plane hurdle hop, and split squat jump) as described in Table 1. During each repetition of the plyometric exercises, subjects landed with their dominant foot on a force plate (Bertec 4060, Columbus, OH). In all jumps but the single-leg sagittal plane hurdle hop, the non-dominant leg also contacted the ground off of the force plate. Movement velocity was controlled during each task via a digital metronome set at 76 bpm with one landing on the force platform occurring at every beat of the metronome. Hurdle hops were performed over a 10.16 cm hurdle. The specific plyometric exercises included in this study were chosen by reviewing the literature and identifying the most common exercises used to alter biomechanical factors associated with ACL loading or ACL injury risk (Arabatzi et al., 2010; Chimera et al., 2004; Hewett et al., 1996; Irmischer et al., 2004; Lephart et al., 2005; Myer et al., 2006a; Myklebust et al., 2003; Pasanen et al., 2008; Steffen et al., 2008; Wilkerson et al., 2004).

After the explanation of each exercise, subjects completed at least two trials to ensure proper performance. Proper performance was defined as the subject maintaining balance throughout the trial, landing with the dominant foot fully on the force plate during each jump, completing the exercise in rhythm with the metronome, and refraining from contact with the hurdle. Following the practice trials, each subject completed three, five-second MVICs for each muscle in a block randomized order. One minute of rest was given between MVICs. Then, subjects completed three sets of each plyometric exercise in a block randomized order. One set of each plyometric exercise included three consecutive repetitions. Three sets and repetitions were chosen to reduce the effects of fatigue while providing enough trials to ensure reliability of EMG data. One minute of rest was given between sets of plyometric exercises. If subjects did not display proper performance during a plyometric exercise set, the trial was discarded and repeated. Subjects were excluded from statistical analysis if they performed 5 or unacceptable trials throughout the testing protocol.

2.3. Data reduction

Data were sampled at 1000 Hz. EMG data during the plyometric exercise and MVIC trials were bandpass (20–350 Hz) and notch (59.5–60.5 Hz) filtered (4th order Butterworth), and smoothed using a 15 ms root mean squared sliding window (Norcross et al., 2010a). Force plate data were lowpass filtered at 60 Hz (4th order Butterworth). Data were reduced using custom LabVIEW software (National Instruments, San Antonio, TX, USA). Mean EMG amplitude of each muscle was calculated for both the preparatory and loading phases of each plyometric task. The preparatory phase was defined as the 150 ms before initial contact (vertical ground reaction force > 10 N) (Lephart et al., 2005). The loading phase was defined as the 100 ms after ground contact. This time frame was chosen to evaluate gluteal and hamstring muscle activation during the time at which the ACL is most susceptible to injury (Koga et al.,

Table 2
Mean EMG values of plyometric exercises for the preparatory phase.

	dlsag	frontal	180	slsag	splitsquat
Lateral hamstrings	24.03 (13.65)	19.86 (19.40)	21.15 (17.22)	51.66 (32.01)[a]	21.90 (13.06)
Medial hamstrings	25.03 (12.95)[b]	18.05 (19.67)	16.97 (11.74)	34.09 (19.71)[a]	22.48 (12.32)[c]
Gluteus medius	29.04 (12.44)	27.72 (11.05)	25.39 (8.72)	43.99 (29.01)[a]	25.16 (12.24)
Gluteus maximus	17.27 (12.30)	16.33 (15.48)	11.74 (7.72)	45.32 (32.84)[a]	14.32 (19.89)

dlsag = double leg sagittal plane hurdle hops; frontal = frontal plane hurdle hops; 180 = 180° hops; slsag = single leg sagittal plane hurdle hops; splitsquat = split squat jumps.
[a] Significantly greater than dlsag, frontal, 180, and splitsquat.
[b] Significantly greater than frontal and 180.
[c] Significantly greater than 180.

2010; Norcross et al., 2010a, 2010b). A total of 3 trials with 3 samples per trial resulted in 9 mean EMG values for each muscle during both landing phase for each plyometric exercise. Then, the data from the 9 repetitions of each phase were averaged together to calculate the total mean EMG amplitude for each muscle during each phase.

Three MVIC values were collected for each muscle. A series of 1 s moving averages were calculated across the 5 s trial, and the largest 1 s mean amplitude was identified as the MVIC mean amplitude (Homan et al., 2012). The mean of the 3 MVIC values for each muscle was then calculated to represent maximal EMG amplitude. The mean EMG amplitude for each muscle during each phase was then divided by the mean MVIC value to normalize the data.

2.4. Statistical analyses

Normalized mean EMG amplitudes were compared across exercises using separate one-way repeated-measures ANOVAs for each muscle group during both phases, resulting in a total of 8 analyses. Since recent studies by Dwyer et al. (2010) and Sigward and Powers (2006) did not note differences in hamstring or gluteal activity between sexes, sex differences were not analyzed. One subject's data were excluded from statistical analysis because she could not complete an acceptable number of trials. Two subjects' lateral and medial hamstring data were excluded from the statistical analysis because the EMG electrodes fell off during testing. Four other subjects' EMG data were excluded from all statistical analyses because they were identified as statistical outliers for at least one muscle (i.e. values ≥ 5SD's above the mean). Significant ANOVA models were evaluated post hoc via Tukey's HSD test. Statistical significance was established a priori $\alpha \leq 0.05$. In addition, we conducted intraclass correlation coefficients (ICC's) across 6 repetitions of each exercise for each muscle to ensure reliability.

3. Results

Mean values for each muscle during each exercise are provided in Tables 2 and 3. Effect sizes are provided in Tables 4 and 5. EMG amplitudes displayed good-to-excellent reliability across trials as evidenced by reliability coefficients ($ICC_{3,k}$) ranging 0.54–0.93 for the preparatory phase and 0.72–0.92 for the loading phase. Only the lateral hamstring activation during the preparatory phase of the frontal plane hurdle hops resulted in a reliability coefficient less than 0.70.

3.1. Medial hamstrings

Significant differences ($P < .001$) between exercises were found for medial hamstring activation during both the preparatory and loading phases (Fig. 1, Tables 2 and 3). Upon post-hoc analysis, the single-leg sagittal plane hurdle hop demonstrated significantly greater preparatory activity than all other exercises and greater loading phase activity than all other exercises except for the double-leg sagittal plane hurdle hop. Additionally, the double-leg sagittal plane hurdle hops displayed greater activation than the 180° jumps.

3.2. Lateral hamstrings

Significant differences ($P < .001$) between exercises were found for lateral hamstring activation during both the preparatory and loading phases (Fig. 1, Tables 2 and 3). Post-hoc analysis determined that single-leg sagittal plane hurdle hop demonstrated significantly greater lateral hamstring activity than all other exercises during the preparatory phase and greater activity than all other exercises except for the double-leg sagittal plane hurdle hop during the loading phase. During the preparatory phase, significantly less lateral hamstring activity was observed for the 180° jumps compared to the double-leg sagittal plane hurdle hops and split squat jumps and the frontal plane hurdle hops compared to the double-leg sagittal hurdle hops. During the loading phase, 180° jumps displayed significantly less activation than all other exercises.

3.3. Gluteus medius

Significant differences ($P < .001$) between exercises were found for gluteus medius activation during both the preparatory and loading phases (Fig. 1, Tables 2 and 3). Upon post-hoc analysis, the single-leg sagittal plane hurdle hop demonstrated significantly greater preparatory and loading activities than all other exercises. Conversely, the 180° jumps displayed significantly less gluteus medius activation than the single-leg sagittal plane hurdle hops, split squat jumps, and double-leg sagittal plane hurdle hops.

Table 3
Mean EMG values of plyometric exercises for the loading phase.

	dlsag	frontal	180	slsag	splitsquat
Lateral hamstrings	52.00 (39.63)[a]	40.63 (42.07)	36.06 (28.83)	57.59 (35.76)[b]	44.05 (31.35)
Medial hamstrings	29.41 (16.57)[a]	24.91 (12.86)[a]	19.39 (12.68)	33.21 (20.46)[b]	27.42 (15.36)[a]
Gluteus medius	60.48 (31.53)[c]	39.56 (18.48)	47.30 (25.18)	100.91 (50.80)[d]	61.76 (37.00)[c]
Gluteus maximus	62.96 (53.58)[e]	39.51 (43.40)	28.16 (33.94)	69.43 (28.61)[b]	47.34 (27.18)

dlsag = double leg sagittal plane hurdle hops; frontal = frontal plane hurdle hops; 180 = 180° hops; slsag = single leg sagittal plane hurdle hops; splitsquat = split squat jumps.
[a] Significantly greater than 180.
[b] Significantly greater than frontal, 180, and splitsquat.
[c] Significantly greater than frontal.
[d] Significantly greater than dlsag, frontal, 180, and splitsquat.
[e] Significantly greater than frontal and 180.

Table 4
Preparatory phase effect sizes.

		dlsag	frontal	180	slsag	splitsquat
dlsag	LH		0.344	0.186	−1.121	0.160
	MH		0.540	0.653	−0.544	0.202
	Gmed		0.112	0.340	−0.897	0.327
	Gmax		0.068	0.538	−1.131	0.254
frontal	LH	−0.344		−0.091	−1.336	−0.173
	MH	−0.540		0.096	−0.929	−0.384
	Gmed	−0.112		0.234	−1.006	0.230
	Gmax	−0.068		0.374	−1.129	0.150
180	LH	−0.186	0.091		−1.187	−0.049
	MH	−0.653	−0.096		−1.055	−0.458
	Gmed	−0.340	−0.234		−1.205	0.023
	Gmax	−0.538	−0.374		−1.407	−0.272
slsag	LH	1.121	1.336	1.187		1.217
	MH	0.544	0.929	1.055		0.707
	Gmed	0.897	1.006	1.205		1.160
	Gmax	1.131	1.129	1.407		1.267
splitsquat	LH	−0.160	0.173	0.049	−1.217	
	MH	−0.202	0.384	0.458	−0.707	
	Gmed	−0.327	−0.230	−0.023	−1.160	
	Gmax	−0.254	−0.150	0.272	−1.267	

dlsag = double leg sagittal plane hurdle hops; frontal = frontal plane hurdle hops; 180 = 180° hops; slsag = single leg sagittal plane hurdle hops; splitsquat = split squat jumps; LH = lateral hamstrings; MH = medial hamstrings; Gmed = gluteus medius; Gmax = gluteus maximus.

3.4. Gluteus maximus

Significant differences ($P < .001$) between exercises were found for gluteus maximus activation during both the preparatory and loading phases (Fig. 1, Tables 2 and 3). Post-hoc analysis determined that the single-leg sagittal plane hurdle hops demonstrated significantly greater preparatory gluteus maximus activity than all other exercises. In the loading phase, the frontal plane hurdle hops and 180° jumps displayed less gluteus maximus activity than the single-leg and double-leg sagittal plane hurdle hops. The split squat jumps also displayed less loading phase gluteus maximus activity than the single-leg sagittal plane hurdle hops.

4. Discussion

The primary findings of our investigation were that the single-leg sagittal plane hurdle hops consistently produced the greatest activation of the gluteus medius, gluteus maximus, medial hamstrings, and lateral hamstrings during the preparatory phase of all the plyometric exercises tested. On the other hand, the 180° jumps consistently produced relatively low levels of activity of the gluteals and hamstrings that did not exceed those of any of the other plyometric exercises.

Gluteus maximus activation during the preparatory and loading phases was greatest during the single leg sagittal plane hurdle hop, and the values observed in our study were only slightly greater than the previously observed gluteus maximus activation values during single leg landings (Zazulak et al., 2005). In the loading phase, gluteus maximus activity during the double leg sagittal plane hurdle hops and split squat jumps was significantly higher than gluteus maximus activation during the frontal plane hurdle hops and split squat jumps. The observation that gluteus maximus activity is greater during sagittal plane exercises may be explained by the fact that the primary action of the gluteus maximus is to extend the hip while the secondary action is to produce external rotation of the femur (Neumann, 2010). These actions of the gluteus maximus would suggest that more muscle fibers are recruited

Table 5
Loading phase effect sizes.

		dlsag	frontal	180	slsag	splitsquat
dlsag	LH		0.278	0.460	−0.148	0.222
	MH		0.302	0.679	−0.204	0.125
	Gmed		0.810	0.462	−0.956	0.254
	Gmax		0.481	0.776	−0.148	0.368
frontal	LH	−0.278		0.124	−0.437	−0.095
	MH	−0.302		0.433	−0.485	−0.176
	Gmed	−0.810		−0.350	−1.605	−0.759
	Gmax	−0.481		0.292	−0.815	−0.216
180	LH	−0.460	−0.124		−0.663	−0.265
	MH	−0.679	−0.433		−0.812	−0.570
	Gmed	−0.462	0.350		−1.337	−0.457
	Gmax	−0.776	−0.292		−1.315	−0.625
slsag	LH	0.148	0.437	0.663		0.403
	MH	0.204	0.485	0.812		0.320
	Gmed	0.956	1.605	1.337		1.106
	Gmax	0.148	0.815	1.315		0.792
splitsquat	LH	−0.222	0.095	0.265	−0.403	
	MH	−0.125	0.176	0.570	−0.320	
	Gmed	−0.254	0.759	0.457	−1.106	
	Gmax	−0.368	0.216	0.625	−0.792	

dlsag = double leg sagittal plane hurdle hops; frontal = frontal plane hurdle hops; 180 = 180° hops; slsag = single leg sagittal plane hurdle hops; splitsquat = split squat jumps; LH = lateral hamstrings; MH = medial hamstrings; Gmed = gluteus medius; Gmax = gluteus maximus.

to perform hip extension than to perform femoral external rotation during this task. Furthermore, eccentric action of the gluteus maximus would be necessary to limit hip flexion moment and anterior pelvic tilt (Alvim et al., 2010). Although not measured in this study, the hip flexion moment for this exercise was likely greater than that for the frontal and transverse plane exercises, thus requiring greater activation of the gluteus maximus.

Activation of the gluteus medius during the loading phase of the frontal plane hurdle hop was significantly less than any of the other exercises except the 180° jumps. This finding is unexpected and contrary to our hypothesis, as a high level of gluteus medius activity would seemingly be necessary during this task to resist frontal plane motion and previous literature has demonstrated that frontal plane step-ups produce greater activation of the gluteus medius than sagittal plane step-ups (Mercer et al., 2009). The difference in gluteus medius activation between our study and other studies may exist because of the data collection process. EMG data were collected only as the subject jumped toward the non-dominant limb. While moving in this direction, subjects may have used their non-dominant limb's hip abductors rather than their dominant limb's hip abductors to stabilize the pelvis in order to change directions. However, this theory cannot be confirmed because no EMG activity was measured on the non-dominant limb. Given that frontal plane hurdle hops seem to be an important component in plyometric programs aimed at ACL injury prevention (Hewett et al., 1999; Mandelbaum et al., 2005) and confounding factors may have been present, further investigation is needed to determine whether frontal plane hurdle hops are an acceptable plyometric exercise to change biomechanics that lead to ACL injury.

Though their primary action occurs in the sagittal plane, the hamstrings also control knee varus and valgus motions (Lloyd et al., 2005). Because the medial hamstrings have a moment arm that can create a knee varus moment (Lloyd et al., 2005), higher activation of the medial hamstrings may limit valgus knee motion and ACL loading (Lloyd et al., 2005; Palmieri-Smith et al., 2008). During this study, the single-leg sagittal plane exercise produced greater preparatory phase activation of the medial hamstrings than all other exercises tested and produced greater loading phase medial hamstring activation than all other exercises tested except for the double-leg sagittal plane exercise. The double-leg sagittal plane exercise produced greater preparatory medial hamstring activation than the frontal plane and transverse plane exercises and greater loading phase activation than the transverse plane exercises. These results suggest that the use of sagittal plane plyometric exercises may be more effective than frontal or transverse plane exercises for protecting against knee valgus and decreasing ACL loading.

Greater lateral hamstring activity during single-leg dynamic exercises compared to double-leg exercises has been documented in previous literature (McCurdy et al., 2010). High levels of lateral hamstring activation may reduce anterior tibial translation and subsequently ACL loading (Krogsgaard et al., 2002) and improve knee joint stability if coupled with medial hamstring activity (Lloyd and Buchanan, 2001). However, lateral hamstring activity may produce valgus loading (Palmieri-Smith et al., 2008). This valgus moment may be mitigated by high medial hamstring activity, which has been shown to reduce peak knee valgus angles (Palmieri-Smith et al., 2008). Since the single-leg sagittal plane hurdle hops also displayed highest medial hamstring activity, the potentially negative effects of the high lateral hamstring activity may be mitigated, and the single-leg sagittal plane hurdle hops may still be an effective exercise to include in plyometric ACL prevention programs if they are completed correctly. Future research is necessary to elucidate the relative contributions of medial and lateral hamstring activities to ACL loading.

The 180° jumps produced either the smallest or 2nd smallest activation of each muscle during both phases of all of the exercises. We expected the gluteus maximus to be more active during the 180° hops to control internal rotation of the hip. However, these results suggest that the 180° jump requires less gluteus maximus activity to control and produce transverse plane movement than to control and produce sagittal plane movement. This observation may have occurred because the primary action of the gluteus maximus is to extend the hip, and the secondary action is to produce external rotation of the femur and control internal rotation eccentrically. Other muscles that cannot be measured with surface EMG because of their depths (e.g. piriformis, obturators, quadratus femoris, and gamelli) may aid the gluteus maximus in controlling femoral internal rotation during the 180° plyometric exercise, thus requiring less gluteus maximus activity (Neumann, 2010). Additionally, subjects completed the majority of transverse plane rotation during this task early during the flight phase. Because the rotation was complete, subjects possessed little rotational momentum when they contacted the ground, leading to limited internal rotation momentum and requiring minimal gluteus maximus/hip external rotator activity. Because muscle activation is an important factor in determining which exercises should be used in the clinical setting, the 180° jumps may not be an important exercise to include in plyometric programs aimed at preventing ACL injury.

This study has several limitations. First, the subjects were recreationally active; thus application to athletes may be limited. Type of exercise is only one component of muscle activation during plyometric exercises. Other factors not examined, such as cueing and jump distance, may also affect muscle activation during plyometric exercises. Other limitations of this study were that data were only collected while the subjects were jumping in one direction, so all conclusions drawn from this study are based on half of the landings performed by each subject. Also, EMG data were only obtained from one limb during double limb exercises, so we could not assess the contributions of the non-dominant limb during the plyometric exercises.

5. Conclusion

The single-leg sagittal and double-leg sagittal plane hurdle hops were the most effective exercises for activating the hamstring and gluteal muscles and may, therefore, be effective exercises to include in ACL injury prevention programs. Single-leg sagittal plane plyometric exercises produced the greatest activation of the gluteus medius, gluteus maximus, and hamstrings while the 180° jumps produce the least activation of these muscles. Therefore, incorporating the single leg sagittal plane hurdle hops and double leg sagittal plane hurdle hops in ACL prevention programs may improve their efficacy while more research on muscle activation during frontal plane tasks is needed to determine their effect. On the other hand, eliminating 180° jumps may improve the efficiency with which ACL plyometric prevention programs are completed without affecting the efficacy. Further research should be performed to determine the long-term effects of muscle activation resulting from plyometric exercises and whether these changes in muscle activation lead to changes in biomechanics related to ACL injury.

Conflicts of interest

There is no conflict of interest.

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